

Five Situations

A walkthrough of the interface, and an honest account of where it strains.

The five situations

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01

Configuring a container card

A Lovelace card that holds schedules filtered by tag — the pattern the existing scheduler card established, with its filter semantics made explicit.

02

A cover, every weekday that is not a holiday or Shabbat

One fixed time, one day set, two exclusion terms — and a term that turns out to remove nothing.

03

Two air conditioners, staged around candle lighting

Sixty minutes before, then a second stage five minutes before that changes mode, temperature and fan and turns on light switches. Four steps.

04

The card as a list, with descriptions

What each schedule is for, in the owner's words, next to what it will do next.

05

One schedule, and its next runs

The enumeration itself — six dated rows, a stated horizon, and a clock change that lands inside the window.

What the screens assume

MODEL / FIXTURE

TWO RULE SHAPES, ONE SCHEDULE

A schedule holds rules, and a rule is either an *At* — a moment, firing service calls — or a *During* — a span, holding desired state. Times are anchors plus offsets, where an anchor is a stable machine key, never display text. Day sets are named objects, not checkboxes buried in a rule.

NOTHING BELOW THE TICK READS A CLOCK

now is threaded as a parameter all the way down. That single constraint is what makes the dry run, the timeline and the live engine the same code path — and it is why a list of future runs is a view rather than a simulation.

18:26

Candle lighting on Friday 18 September 2026, the moment every screen is drawn against. Sunset less eighteen minutes.

32°05'N

Sunsets computed by the standard solar algorithm; Hebrew dates by the fixed arithmetic calendar. Checked against three independent anchors.

0

Invented dates. The first draft used hand-written ones and was wrong by a day throughout — every screen individually plausible.

01

Configuring a container card

The filter is the whole feature. Upstream ships one and never states what it means — so the first job was to read it, not to redesign it.

The card editor, with the filter made explicit



HOME ASSISTANT - CARD EDITOR, LIGHT THEME

PLATE 01

- a **Match any / match all is a stored mode.** Upstream matching is an OR with no alternative and no field to record intent, so “tagged both sanctuary and kiddush” cannot be expressed at all.
- b **Exclusion is promoted out of YAML.** The option already exists upstream and is honoured at runtime — it is simply absent from the editor, so only readers of the docs know it is there.
- c **On and off stop being tags.** Upstream reserves three words inside the same list the real tags come from, with no escape syntax. A schedule genuinely tagged with one of them cannot be filtered for.
- d **A tag that matches nothing says so.** The typo on this screen is deliberate. Free text plus OR matching means a mistyped tag yields a quieter card and no error — which reads as breakage, not as a typo.
- e **The preview carries a count.** Three of fourteen match, one hidden by exclusion. A filter is the one control whose result you cannot infer from its inputs.

02

A cover, on ordinary days

Written exactly as asked, the rule contains a term that does nothing.
The dangerous part is that it is correct.

Every weekday that is not a holiday or Shabbat

A new schedule dialog for raising a cover at 07:15, with a day set of Sunday to Friday and two exclusion terms, one tagged as removing nothing.

NEW SCHEDULE — DAY SET AND EXCLUSIONS

PLATE 02

- a **The day set is a named object.** Not a row of weekday checkboxes buried inside the rule — a thing with a name, editable once and referenced by every schedule that uses it.
- b **One level of set algebra.** Start from a set, then union, intersect or subtract — once. Deeper nesting was refused deliberately; it is the point at which a rule stops being readable aloud.
- c **“Except Shabbat” removes nothing here,** and the term says so on its own row. Sunday to Friday never contained Saturday, and 07:15 on a Friday is hours before candle lighting.
- d **Both buttons are offered.** Remove it, or keep it anyway — keeping a defensive term against a future edit is a legitimate choice, and the interface should not assume otherwise.
- e **Eleven of the next fourteen days.** The count is the check. Three days are grey, and each is grey for a reason you can read.

03

Two stages, before lighting

Four steps. The anchor is not a clock time, the actions travel by two different routes, and the first run turns out to be a Sunday.

Step one – the time is not a time

Step one of the new schedule wizard. Choosing candle lighting as the anchor with a sixty minute negative offset.

STEP 1 – ANCHOR AND OFFSET

PLATE 03

- a **Anchor plus offset, not a clock.** Sixty minutes before candle lighting is a different wall-clock time every week, and the rule stores the relationship rather than the result.
- b **The anchor list is wider than upstream’s.** Upstream whitelists sunrise and sunset only — excluding dawn, dusk and noon, which the sun entity already publishes.
- c **Entity anchors are the reason this project exists.** They were proposed upstream and explicitly refused by the maintainer, which is what makes this a rebuild rather than a contribution.
- d **No magnitude limit.** Upstream’s card caps offsets at four hours; the number has been changed four times since 2020 and never with a stated reason. The engine has no check at all.
- e **Horizon is declared next to the anchor.** Candle lighting is pure local computation, so it is unbounded. An anchor read from a calendar entity is not, and says where it stops.

Step two – one unit per route, side by side

 Step two: two air conditioner actions, one delivered as climate service calls and one as a script, each showing its own failure mode.

STEP 2 – ACTIONS, CLIMATE AND SCRIPT ROUTES

PLATE 04

- a **You said either route was fine.** They are — but they fail differently, so the route sits in the row header rather than behind a settings page. It changes what the row can promise.
- b **Temperature and fan is two calls.** Home Assistant has no atomic set-full-state for climate: setting temperature accepts an optional mode, and nothing accepts fan mode alongside anything else.
- c **So a row can half-succeed,** and results are reported per attribute rather than per row. A row that says “done” when one of its two calls was refused is the failure this avoids.
- d **The script route pre-flights.** A script already running in its default mode logs a line and returns without raising — the caller cannot tell that from success, so the row checks first and records it as skipped.
- e **Amber, here, means one thing:** this can silently not happen. Failure is red, deliberately inactive is dimmed. Amber is reserved for the case where the system reports success and you are still wrong.

Step three – the second stage, five minutes before

Step three: a second stage five minutes before candle lighting turning on light switches and changing the air conditioners to a quieter mode.

STEP 3 – SECOND STAGE

PLATE 05

- a **Two stages, one anchor.** Both are measured from candle lighting — sixty minutes and five minutes — so they keep their spacing as the anchor moves through the year.
- b **Four light switches and two units in one stage.** Upstream forbids this: its actions list may hold only a single service and data combination, which is what forces the script-per-permutation workaround.
- c **The change is to mode, temperature and fan.** Which is three attributes across two calls per unit — the same constraint as stage one, now with more to lose.
- d **Nothing is scheduled to undo it.** That is deliberate and visible: the span ends thirty minutes after havdalah, and what happens at the end is a property of the rule, not an afterthought.
- e **This is where the model and the user diverge.** One preparation, two rules. Hold that thought — it returns in the critique as the most serious thing on these screens.

Step four — and the first run is a Sunday

Step four, the review step showing the first run on Sunday 20 September, erev Yom Kippur, followed by three subsequent Fridays at different times.

STEP 4 — REVIEW AND FIRST RUN

PLATE 06

- a **Sunday 20 September, not Friday.** Erev Yom Kippur. Nobody arranged this — it is what the enumeration returned once the fixture was computed rather than written by hand.
- b **The claim pays for itself.** Every earlier draft asserted in prose that candle lighting is not always a Friday. This step demonstrates it without a sentence of explanation.
- c **Then 17:17, then 17:07, then 16:58.** The times drift because the anchor drifts. A review step showing one fixed time would be hiding the behaviour that matters most.
- d **Havdalah is two days out.** Yom Kippur ends Monday night, so the span runs Sunday evening to Monday evening — and the review says so rather than quietly printing a longer bar.
- e **Review is the same code path as the engine.** Not a preview built separately and liable to drift — the same enumeration, with a different value of now.

04

The list, with descriptions

Drawing it established that the description is not decoration on the row. It is the reason to look at the card at all.

Eight schedules, and what each one is for

A dashboard card listing eight schedules with descriptions and next run times, alongside a sun card showing sunset, candle lighting and havdalah.

CONTAINER CARD – LIST WITH DESCRIPTIONS

PLATE 07

- a

A column of names is a directory. The description is what says something the name does not, and it is why this card is worth the space it takes.
- b

It needs roughly 460 pixels. The default masonry column is 340, at which every description loses its second half — and a truncated reason is worse than none, because it reads as complete.
- c

Next run sits opposite the description. What it is for, and what it will do next, on one line. Sorted by what happens soonest, with anything wrong lifted out of the sort.
- d

The sun card is the fixture, visible. Sunset 18:44, candle lighting 18:26, havdalah 19:33 tomorrow — the same numbers every other screen is derived from.
- e

One row is amber, and it is the wrong row. Kiddush setup was skipped last Friday — and the status has taken the description’s place. That is the second finding in the critique.

05

One schedule, its next runs

The enumeration, printed. Which is the whole premise of the project,
and therefore the screen with the least room to be wrong.

Six dated rows, a stated horizon, one clock change

A schedule detail card listing its next six runs with start and end times, a horizon footer, and a note about the clock change in late October.

NEXT RUNS — WITH HORIZON FOOTER

PLATE 08

- a **Every row is start and end.** Forty-five minutes before candle lighting, thirty minutes after havdalah — and both ends move, so both are printed rather than one plus a duration.
- b **The horizon lives in the footer.** On a list, the horizon is a property of the end of the list, not of the picker. Rows printed past it look exactly like rows before it.
- c **Unbounded gets its own wording.** Silence is indistinguishable from “we did not check”, so a schedule that can be enumerated indefinitely says so in as many words.
- d **16:57, then 15:50.** The clocks go back on 25 October, so the start jumps an hour and seven minutes earlier — correct, and indistinguishable from a bug unless the card says which.
- e **That footer was wrong once.** It quoted candle lighting instead of the start — a real number for that date, in a sentence that parsed. A footer must quote the same quantity as the rows above it.

Where this would strain

Seven pitfalls, ranked. Three of them are severe enough to change the model rather than the interface, and one is a promise the engine cannot keep.

A pitfall is not a bug. It is a place where the interface *will be believed* and should not be.

Ranks below were settled with the owner on 29 September 2026. Two came down; one was re-framed after checking it against the design doc and the prototype, and the original reading of it is recorded on slide 21 as wrong.

↑ SEVERE

Changes the model, not the screen. Shipping without deciding makes it more expensive later, not less.

— MODERATE

Real, and addressable in the interface. Costly if ignored through the first release.

↓ KNOWN

Already recorded against a decision. Listed so the ranking is honest about what is and is not news.

↓ DEFERRED

Accepted as real, then ruled to a later phase. What the interface says in the meantime still has to change.

Three that were severe; one still is

NO.	PITFALL	WHY IT BITES	RANK
01	One occasion, never drawn whole	Getting the place ready for Shabbat is a single thing to the person who wants it, and the model already agrees — recurrence sits on the schedule, the schedule owns the switch and the sensor, and the list and the timeline each give it one row. The defect is that no surface draws the occasion <i>with its parts</i> : authoring walks four steps one stage at a time, and every read surface collapses the evening back to a single line. Disable one stage and nothing says the schedule is half-armed.	↑ SEVERE
02	The description drifts, and is overwritten	The description is free text and nothing ties it to the rule it describes — edit the rule and the sentence silently becomes a lie. Ruled <i>moderate</i> : most users never write one, and it is cheap to change later. The sharper half survives the downgrade — the list card renders skipped-status text <i>into</i> the description slot, so the row that most needs explaining loses its explanation.	– MODERATE
03	The script pre-flight is a race	The row promises it will check a script’s mode before calling and record a skip rather than a false success. But the check and the call are separate operations, and the script can start in between. Ruled to a <i>later phase</i> : the race is rare and closing it is engine work. What changes now is the wording — the interface must describe a check, never a guarantee.	↓ DEFERRED

Four more, in descending order

NO.	PITFALL	WHY IT BITES	RANK
04	Enumeration at scale is unproven	Forty schedules have been drawn as a sorted list, and one schedule has been drawn as a timeline. Forty schedules <i>on</i> a timeline has not been drawn at all — and that view is the product’s entire thesis. The density that makes a single schedule legible is the density that makes forty unreadable.	– MODERATE
05	Anchors have no shallow path	Before a user can schedule anything they must absorb anchor, offset and horizon. The concepts are taught well, but they are taught <i>first</i> , and the thing being replaced lets you pick 07:15 and stop. A product whose simplest task is harder than the incumbent’s loses users before its advantages appear.	– MODERATE
06	Amber must be learned to be read	Red is failure, amber is silent no-op, dim is deliberately inactive — a good contract that a user can only infer by meeting all three. Text accompanies each state today, which is what keeps it accessible; the risk is that the text is treated as optional later, at which point the distinction rests on two hues that a meaningful share of users cannot reliably separate.	– MODERATE
07	The card wants a column it will not get	Descriptions need about 460 pixels; the default masonry column is 340. The fork is already recorded against the card decision, and screen 13 makes it the user’s choice — but shipping a default that truncates the payload is still shipping a default that truncates the payload.	↓ KNOWN

Fix the first one — but not in the schema. The container is already there. What is missing is a view of the occasion with its parts.

This deck first read it as a schema split. That was wrong: the schedule already is the container, in storage and in the entity registry alike. The other six are fixable where they appear — a footer, a column width, a colour contract, a settings default. This one is fixable in exactly two places, and both are early. The editor's shape is what teaches the model; the row's shape is what reminds you of it.

Twenty screens, seven pitfalls

SCREENS

Twenty static screens in the prototype; the five requested situations are the last eight of them.

CAPTURES

Forty-five, light and dark, desktop and phone. The eight plates here are the light desktop set.

FIXTURE

Friday 18 September 2026, early evening. Every date computed; screens one to twelve still carry the earlier hand-written one.

FINDINGS

Eighteen recorded against the design decisions. Seven pitfalls above; the remainder are what worked.